

Basic Civil Engineering Notes

1. Building Materials

- Cement – Types: OPC, PPC, Rapid Hardening, Low Heat.
- Concrete – Mix ratio, curing methods, strength testing.
- Steel – Properties, reinforcement types, corrosion protection.
- Bricks & Blocks – Types, tests, and uses.

2. Surveying

- Chain Surveying, Compass Surveying, Leveling.
- Theodolite – Measuring horizontal & vertical angles.
- Total Station & GPS – Modern surveying tools.

3. Structural Engineering

- Load Types – Dead load, Live load, Wind load, Earthquake load.
- Beams, Columns, Slabs – Basic concepts.
- Foundation Types – Shallow & Deep foundations.

4. Transportation Engineering

- Road Types – Flexible pavement, Rigid pavement.
- Traffic Control – Signals, Signs, Road markings.
- Highway Materials – Bitumen, Aggregates.

5. Environmental Engineering

- Water Treatment – Filtration, Chlorination.
- Wastewater Treatment – Primary, Secondary treatment.
- Solid Waste Management – Collection & Disposal.

6. Construction Management

- Project Planning – CPM & PERT.
- Estimation & Costing – BOQ, Rate analysis.
- Site Safety – PPE, Hazard control.